

IMPLEMENTING DEPENDENCY INJECTION

BookService.java (Updated):

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }
    public void display() {
        System.out.println("BookService is working.");
        bookRepository.display();
    }
}
```

applicationContext.xml (Updated):

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>

    <bean id="bookService"
        class="com.library.service.BookService">
        <property name="bookRepository" ref="bookRepository"/>
    </bean>
</beans>
```

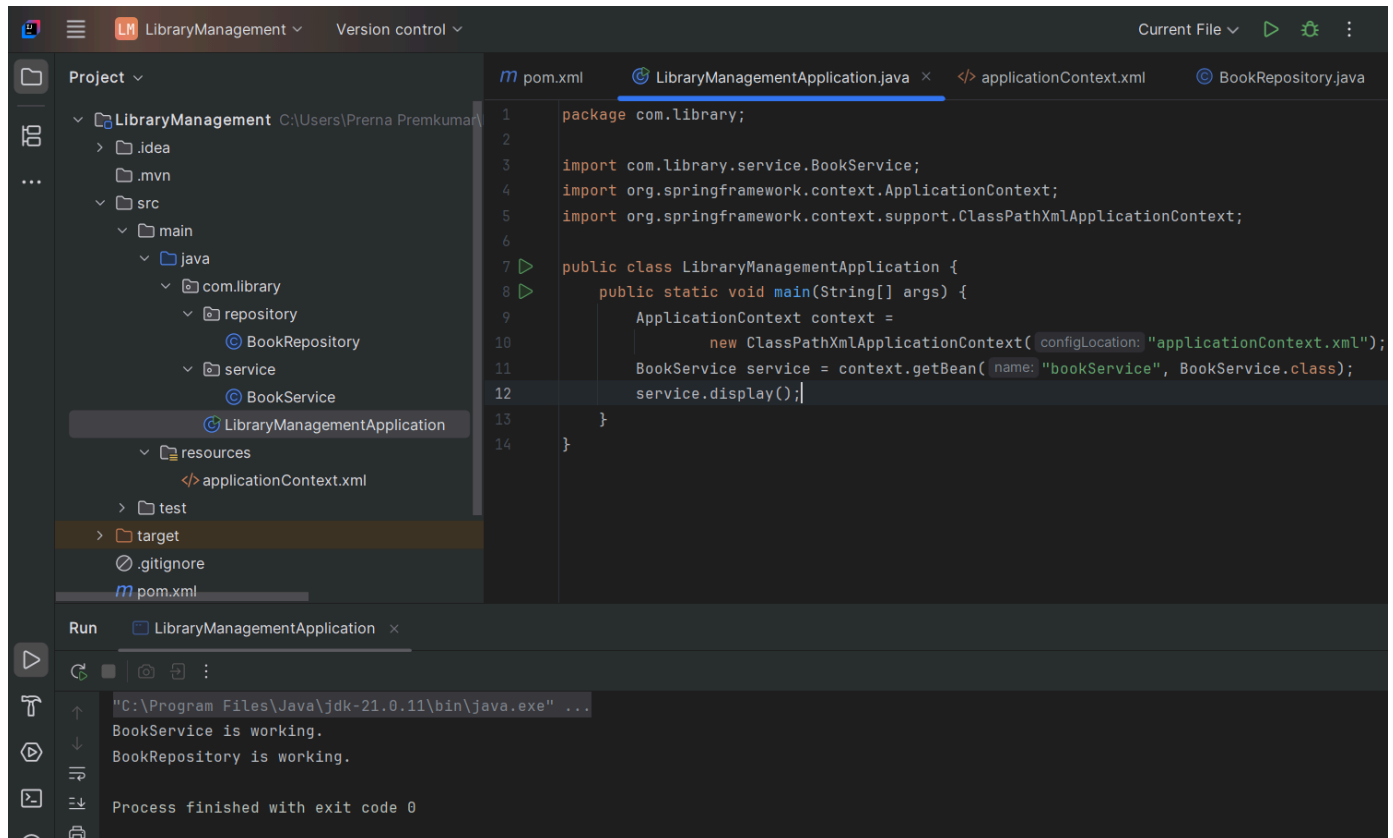
LibraryManagementApplication:

Same as exercise 1.

BookRepository:

Same as exercise 1.

Output:



The screenshot shows an IDE with the following components:

- Project View:** A tree structure on the left showing the project hierarchy. The `LibraryManagementApplication` class is highlighted under the `com.library` package.
- Code Editor:** The `LibraryManagementApplication.java` file is open, showing the following code:

```
1 package com.library;  
2  
3 import com.library.service.BookService;  
4 import org.springframework.context.ApplicationContext;  
5 import org.springframework.context.support.ClassPathXmlApplicationContext;  
6  
7 public class LibraryManagementApplication {  
8     public static void main(String[] args) {  
9         ApplicationContext context =  
10             new ClassPathXmlApplicationContext("applicationContext.xml");  
11         BookService service = context.getBean("bookService", BookService.class);  
12         service.display();  
13     }  
14 }
```
- Run Console:** The output of the application is displayed at the bottom, showing the execution path and the messages:

```
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...  
BookService is working.  
BookRepository is working.  
Process finished with exit code 0
```